MD KARIMUL ISLAM

Software Engineer | AI/ML Engineer | Computer Vision

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INTRODUCTION

Software Engineer building full-stack products, data pipelines and LLM systems, with seven end-to-end systems shipped. Production experience with Next.js / TypeScript, FastAPI, PostgreSQL pipelines, AI APIs and multi-agent automation.

HANDS-ON EXPERIENCE

Founder & Developer | Study Abroad Finder (studyabroadproject.vercel.app) | Remote *Oct 2026 – Present*

- Built and run a full-stack study-abroad platform (Next.js, TypeScript, PostgreSQL): 82,000+ master's programmes at 30,000+ universities from 51 official registers; a nightly LLM pipeline publishes requirements only when the official page's quote backs them.
- Shipped a multi-tenant consultancy workspace (Google OAuth, saved candidates, partner universities, instant match reports, PDF export) with 474 automated tests, scheduled GitHub Actions jobs and weekly backups.

Internship — Oleyn PTE Ltd. (Oleyn.ai) | Prompt Engineering Intern | Remote *Jan – Apr 2026*

- Generated business leads and pitched AI chatbot solutions to prospective clients (including Rangs eMart) using OpenRouter and OpenAI APIs.
- Built and configured conversational bots end to end — LLM settings, knowledge base, prompt sections, and escalation rules — for real business use cases.
- Contributed to an omnichannel customer-engagement platform unifying Facebook, Instagram, TikTok, Gmail, website chat, Telegram, and WhatsApp into a single inbox.
- Configured multi-agent handling — multiple human agents authorized to reply, private agent-to-agent handoff, and manual AI on/off per conversation.
- Set up automatic capture of customer name, number, and other details into a structured dataset.
- **Learned:** Prompt design for both lead-generation and support use cases, and how to configure a full omnichannel bot stack (LLM settings, knowledge base, social/WhatsApp integration, analytics) from the ground up.

Project Lead | GeoChain | Rajshahi *Apr – Sep 2025*

- Built a blockchain-ready property marketplace with identity verification workflows; secured a 2nd UIHP grant (ICT Division & World Bank-backed, two-time grantee in 12 months).

Project Lead | VetConnect | Rajshahi *Oct 2024 – Mar 2025*

- Launched Bangladesh's first digital veterinary platform — GPS-based vet discovery, telemedicine, and an e-commerce prototype — in 6 months, UIHP-funded (ICT Division & World Bank).

SKILLS

- **Programming Languages:** Python, TypeScript, C++, C, SQL
- **Databases:** MySQL, PostgreSQL (Drizzle ORM), SQLite
- **API Integration:** REST API design & consumption (OpenRouter, OpenAI, FastAPI services)
- **Data Structures & Algorithms:** Trees, Graphs, Priority Queues, Dynamic Programming, BFS/DFS
- **Data Preprocessing:** Homology-aware clustering (MMseqs2), sequence-window modelling, data augmentation & normalization
- **ML / DL:** PyTorch, TensorFlow / Keras, scikit-learn
- **Architectures:** CNNs, DenseNet, ResNet, MobileNet, EfficientNet, 1D-CNN
- **CV / XAI:** OpenCV, MediaPipe, Grad-CAM, Integrated Gradients, LIME, ONNX
- **LLM / GenAI:** Prompt Design, Chain-of-Thought, RAG, AI Agents, LLM Evaluation
- **Tools / Serving:** VS Code, Git, GitHub Actions, Next.js, FastAPI, Docker, Linux, Streamlit, Tesseract OCR

EDUCATION

B.Sc. in Computer Science & Engineering | Varendra University, Rajshahi *Sep 2022 – Aug 2026*

- CGPA 3.90 / 4.00 · Merit Scholarship (ranked 13th of 193 in cohort)
- Hult Prize Campus Team Leader (2026)

PUBLICATIONS

- **MEFNet** — Hybrid CNN for Waste Classification + XAI, IEEE PECCII 2026 (First Author) — 97.71% accuracy, outperformed MobileNetV2/V3, EfficientNetB0, and ResNet50V2 (competitive with heavier ConvNeXt-Tiny at 97.94%).
- **FusionNet** — Deepfake Detection Framework, VIJIR Vol. 1(2), 2025 (First Author) — extended from a UCICS 2025 conference paper — 91.22% accuracy, AUC 0.98, F1 92.56%, FGSM-tested.
- **LitePhospho** — XAI CNN for Phosphorylation Prediction, Undergraduate Thesis, Completed & Presented (First Author) — manuscript in preparation for IEEE TCBB; homology-aware benchmarking, Integrated Gradients.
- **Enhanced MobileNetV1** for Early Autism Detection, IEEE QPAIN 2025 (Co-Author).

PROJECTS ARCHIVE

- **AI Job Tracker —** full-stack SaaS: FastAPI + PostgreSQL with JWT auth, a Postgres job queue and 128 integration tests; Next.js frontend.
- **Agentic Research Assistant —** RAG-based Q&A over research papers with page-level citations, self-scored for faithfulness (FAISS, Groq).
- **Multimodal Visual Analyst —** vision-language app for image Q&A and structured JSON extraction (Llama 4 Scout).
- **Audio Intelligence —** turns recordings into transcripts, structured reports, and a Q&A chat (Whisper + LLM).
- **Churn Prediction (MLOps) —** scikit-learn model served with FastAPI with a live dashboard, Docker, and CI.
- **LitePhospho —** explainable phosphosite predictor served via ONNX with a live Streamlit demo.

ACHIEVEMENTS

- First-author publications: VIJIR 2025 and IEEE PECCII 2026 (presented).
- Graduated August 2026 with CGPA 3.90/4.00; Merit Scholarship recipient, ranked 13th of 193 in cohort (2022–2026).
- 2× UIHP Innovation Program grantee (ICT Division & World Bank) — funding for GeoChain and VetConnect.

CERTIFICATIONS & LANGUAGES

- Google AI Agents Intensive (5-Day) · Machine Learning with Python · Data Visualization with Python · Google Analytics
- Bengali — Native · English — Professional working proficiency